

YAMAGUCHI, Japan

WMO: 477840

Lat: 34.1625N

Lon: 131.4622E

Elev: 20

StdP: 101.09

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.1	(c) -1.1	(d) -8.2	(e) 1.9	(f) 4.4	(g) -6.9	(h) 2.1	(i) 4.4	(j) 7.1	(k) 5.0	(l) 6.3	(m) 5.3	(n) 1.4	(o) 60	(p) 0.364

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 7.9	(c) 34.2	(d) 25.6	(e) 33.0	(f) 25.2	(g) 31.9	(h) 24.7	(i) 26.5	(j) 32.0	(k) 26.0	(l) 30.9	(m) 25.5	(n) 30.0	(o) 3.2	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.2	20.3	28.7	24.6	19.7	28.4	24.1	19.1	28.1	83.0	32.1	80.5	31.0	78.3	30.1	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.5	(b) 5.4	(c) 4.8	(e) -4.0	(f) 35.8	(g) 1.4	(h) 1.2	(i) -5.1	(j) 36.7	(k) -5.9	(l) 37.3	(m) -6.7	(n) 38.0	(o) -7.8	(p) 38.9		
(a) 6.5	(b) 5.4	(c) 4.8	(e) -4.7	(f) 27.2	(g) 1.4	(h) 0.9	(i) -5.7	(j) 27.8	(k) -6.6	(l) 28.4	(m) -7.3	(n) 28.8	(o) -8.3	(p) 29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	4.8	5.8	9.1	14.3	19.2	22.9	27.0	28.1	24.2	18.4	12.3	7.0		
	DBStd	8.47	2.40	2.93	3.11	3.10	2.23	2.02	2.25	1.85	2.70	2.88	3.30	2.91		
	HDD10.0	458	162	121	56	5	0	0	0	0	0	0	14	101		
	HDD18.3	1766	420	349	287	125	16	0	0	0	0	34	182	351		
	CDD10.0	2704	1	4	27	132	286	388	528	562	426	259	83	8		
	CDD18.3	971	0	0	0	3	44	138	269	304	177	36	1	0		
	CDH23.3	8589	0	0	0	36	320	813	2633	3302	1316	167	2	0		
	CDH26.7	3033	0	0	0	3	54	186	1016	1382	381	12	0	0		
(14) Wind	WSAvg	1.7	1.6	1.7	1.9	2.0	2.0	1.7	1.9	1.9	1.7	1.5	1.4	1.5		
(15) Precipitation	PrecAvg	1823	80	84	131	155	187	266	305	189	171	89	92	75		
	PrecMax	2529	176	136	249	327	386	449	705	528	401	230	230	143		
	PrecMin	1229	27	26	58	47	63	54	54	43	43	9	27	11		
	PrecStd	335	38	28	51	66	76	120	190	115	87	60	52	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.5	17.7	21.2	26.7	30.4	31.6	35.7	36.0	33.0	28.2	23.1	17.3		
		MCWB	10.7	12.9	13.3	16.8	18.3	23.0	26.4	26.3	24.8	21.7	17.0	13.3		
	2%	DB	12.6	15.3	18.9	24.2	28.1	30.0	34.0	34.6	31.5	26.5	21.2	15.4		
		MCWB	8.9	10.8	12.3	15.1	17.9	21.6	25.5	25.8	23.8	20.0	16.0	11.3		
	5%	DB	11.1	13.3	17.0	22.2	26.6	28.8	32.9	33.5	30.2	25.2	19.8	14.0		
		MCWB	7.4	9.1	11.6	14.2	17.0	21.1	25.3	25.4	23.3	19.0	15.2	10.2		
	10%	DB	9.7	11.5	15.4	20.5	25.1	27.5	31.6	32.4	28.9	23.9	18.3	12.5		
		MCWB	6.2	7.8	10.8	14.2	16.7	20.7	24.9	25.0	22.6	18.2	14.3	9.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	14.7	15.6	19.6	21.7	25.0	27.2	27.4	25.9	23.7	19.2	14.7		
		MCDB	13.6	16.9	17.4	22.2	24.7	28.6	33.3	33.5	30.7	26.3	20.8	15.7		
	2%	WB	9.7	12.0	14.2	17.7	20.6	24.1	26.5	26.7	25.0	22.2	17.7	12.5		
		MCDB	11.3	14.0	16.8	20.9	23.5	27.3	32.0	32.4	29.0	24.8	19.9	14.2		
	5%	WB	8.1	10.0	12.9	16.5	19.5	23.4	25.9	26.2	24.3	20.6	16.3	11.0		
		MCDB	10.1	12.4	15.9	20.0	23.1	26.2	31.1	31.1	28.3	23.6	18.7	13.0		
	10%	WB	6.8	8.3	11.3	15.4	18.6	22.5	25.4	25.7	23.8	19.1	14.9	9.6		
		MCDB	9.2	11.0	14.6	18.9	22.7	25.5	30.0	30.3	27.5	22.3	17.8	12.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	8.9	10.1	10.7	10.5	8.1	7.4	7.9	8.2	9.6	9.6	8.7		
		MCDBR	10.6	11.6	13.2	14.2	14.1	10.8	9.7	9.6	9.2	10.8	10.7	10.7		
	5% WB	MCWBR	7.0	7.4	7.7	6.9	6.0	4.3	3.3	3.2	3.4	5.0	6.1	7.0		
		MCWBR	9.4	9.2	10.0	9.6	8.8	6.7	8.8	8.6	7.4	7.5	8.4	9.2		
(40) Clear-Sky Solar Irradiance	taub	taub	0.442	0.486	0.555	0.582	0.582	0.596	0.596	0.555	0.515	0.469	0.444	0.433		
		taud	2.012	1.932	1.783	1.773	1.829	1.878	1.923	2.033	2.090	2.112	2.094	2.047		
	Edn at Noon	Edn at Noon	727	740	723	730	736	724	720	743	752	752	723	708		
		Edh at Noon	138	166	209	221	211	201	191	168	152	137	127	127		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.05	2.79	3.87	4.78	5.26	4.57	4.85	5.09	4.02	3.38	2.47	1.90		
	RadStd	RadStd	0.18	0.25	0.29	0.41	0.59	0.47	0.72	0.55	0.41	0.35	0.20	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+46
(47) Regional (29 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page